LEVEL UP ACADEMY

Class 10 | GSEB | Science

Ch 5 Test (GSEB pattern): Life Processes

Chapter / Topic: Life Processes

Name: _____ Roll No.: _____ Date: _____

Time: 50 minutes **Max. Marks: 25**

General instructions:

1. All questions are compulsory.
2. Section A: objective questions, 1 mark each. Section B: 2 marks each. Section C: 3 marks each. Section D: 4 marks.
3. Write balanced equations and draw neat, labelled diagrams wherever needed.

SECTION A

9 marks

Choose the correct option.

Q1. The kidneys in human beings are a part of the system for: **[1]**

(a) excretion (b) nutrition
(c) respiration (d) transportation

Q2. Which of the following shows holozoic nutrition? **[1]**

(a) Cuscuta (b) Mushroom
(c) Amoeba (d) Green plants

Q3. The xylem in plants is responsible for the transport of: **[1]**

(a) food (b) water
(c) amino acids (d) oxygen

Q4. The finger-like projections on the inner lining of the small intestine are called: **[1]**

(a) alveoli (b) nephrons
(c) villi (d) stomata

Q5. The autotrophic mode of nutrition requires: **[1]**

(a) carbon dioxide and water (b) chlorophyll
(c) sunlight (d) all of the above

Q6. Which blood vessel carries deoxygenated blood from the heart to the lungs? **[1]**

(a) Pulmonary artery (b) Pulmonary vein
(c) Aorta (d) Vena cava

Q7. In aerobic respiration, the breakdown of pyruvate to give carbon dioxide, water and energy takes place in the: **[1]**

(a) cytoplasm (b) chloroplast
(c) nucleus (d) mitochondria

Q8. The opening and closing of stomata is caused by: **[1]**

(a) movement of the epidermis
(b) a change in the water content of the guard cells
(c) pressure in the xylem
(d) flow of food in the phloem

Q9. Which blood vessel carries oxygenated blood from the lungs to the heart? **[1]**

(a) Pulmonary vein (b) Pulmonary artery
(c) Aorta (d) Vena cava

SECTION B

6 marks

Q10. How are the alveoli designed to maximise the exchange of gases? **[2]**

Q11. Why do aquatic organisms breathe much faster than terrestrial organisms? **[2]**

Q12. Why is it necessary to separate oxygenated and deoxygenated blood in mammals and birds? **[2]**

SECTION C

6 marks

Q13. Answer the following about photosynthesis: **[3]**

- **(i)** Write the balanced equation for photosynthesis. **[1]**
- **(ii)** List the three events that occur during photosynthesis. **[2]**

Q14. Answer the following about the human heart: **[3]**

- **(i)** How many chambers does it have? **[1]**
- **(ii)** Which chamber pumps oxygenated blood to the rest of the body? **[1]**
- **(iii)** What is the function of the valves in the heart? **[1]**

SECTION D

4 marks

Read the passage and answer the questions that follow.

Q15. When the kidneys fail, an artificial kidney (dialysis machine) can be used. It contains a number of tubes with a semi-permeable lining, suspended in a tank filled with dialysing fluid. This fluid has the same osmotic pressure as blood, except that it contains no nitrogenous wastes. The patient's blood is passed through these tubes, and the nitrogenous wastes pass from the blood into the dialysing fluid. The purified blood is pumped back into the patient. **[4]**

- **(i)** Name the main nitrogenous waste removed from the blood. **[1]**
- **(ii)** Why does the dialysing fluid contain no nitrogenous wastes? **[1]**
- **(iii)** How is dialysis different from the working of a healthy kidney? **[2]**

~ All the best! ~